SpatialNet: one forward pass from a glasses frame to the pointed target

Shared trunk



Input frame, RGB 3×160×160

ShuffleNetV2 0.5x

P1 48×20×20 · P2 96×10×10 · P3 192×5×5

ImageNet init · 143K params

BiFPN ×2

192×20×20

fast normalised fusion + DW3×3 + SimAM · 81K

CoordConv + reduce

256×20×20

+x,y maps → 194 ch
DW3×3 + PW → 256 · 52K

ConvMixer ×2

DW5×5 + PW, residual · 146K

S

Shared map S · 256×20×20 read by all four heads

1 Wearer hand

104K



heatmap 20×20 (DW3×3 → PW1×1)

local soft-argmax, r=2 → centre c_h

3×3 bilinear sample → MLP → (w, h)

RoIAlign 5×5 on 1.15·b_h → descriptor v_h (128-d)

hand box b_h

c_h, b_h

2 Fingertip

52K



heatmap + learned term on offsets (Δx, Δy) to c_h

+ log-prior: tip stays inside 1.3·b_h

local soft-argmax, r=1 → fingertip p

3×3 sample → descriptor v_p (128-d)

fingertip p

v_h, v_p

3 Pointing direction

33K



MLP[v_h, v_p]: 256 → 128 → 2

L2-normalised (sin θ, cos θ)

unit ray θ

p, θ

4 Pointed target

60K



6 ray maps from (p, θ): Δx, Δy, t along ray, |s| across ray, cos θ, sin θ

geometry logits (zero-init) + visual logits

local soft-argmax, r=2 → c_t

3×3 sample → (w, h)

target box b_t

Presence gates

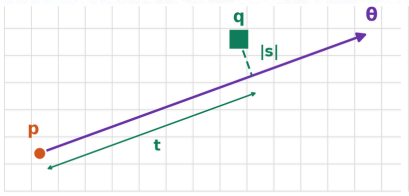
hand gate MLP[GAP(S), peak$_h$, v_h] · 25K

pointing gate MLP[v_h] · 8K

target gate MLP[v_h, v_p, peak$_t$] · 17K

→ **cascade: hand → pointing → target ⇒ scene (H P T)**

Ray coordinates



per cell q: Δx, Δy, t, |s|, cos θ, sin θ

Training only

anchor forcing: ground-truth anchors, prob. 0.5 → 0 in 20 epochs

distractor-hand copy-paste

ownership-weighted heatmap negatives (×5 at pasted hands)

722,104 parameters · 0.23 GFLOPs · 1.6 MB (FP16) · one forward pass